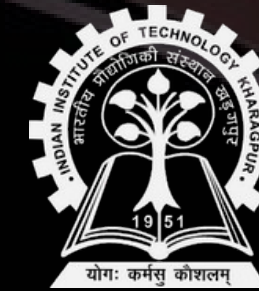


THE
FiSTA



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VajraGrid

AI-Driven Cyber Defense for Smart Power Grids

AMD Slingshot 2026 • AI + Cybersecurity & Privacy

Team Innovibe

D Shantan Dheer (Lead)

MJ Emmanuel • Jakkula Tarun



PROBLEM STATEMENT



India's 575+ GW grid is digitizing fast — 20 Cr+ smart meters create a massive attack surface with catastrophic consequences.

₹100+ Cr

Downtime Cost/hr

20 Million

Affected Mumbai '20

30-90 min

Manual Response

ZERO

AI Grid Defense

⚠ **Real-World Incidents**

- 2020 Mumbai: RedEcho malware in SCADA
20M people lost power
- Ukraine 2015/16: Nation-state grid attacks
caused cascading blackouts
- India 2023: Ransomware on utilities
zero automated response

● **Critical Gaps Today**

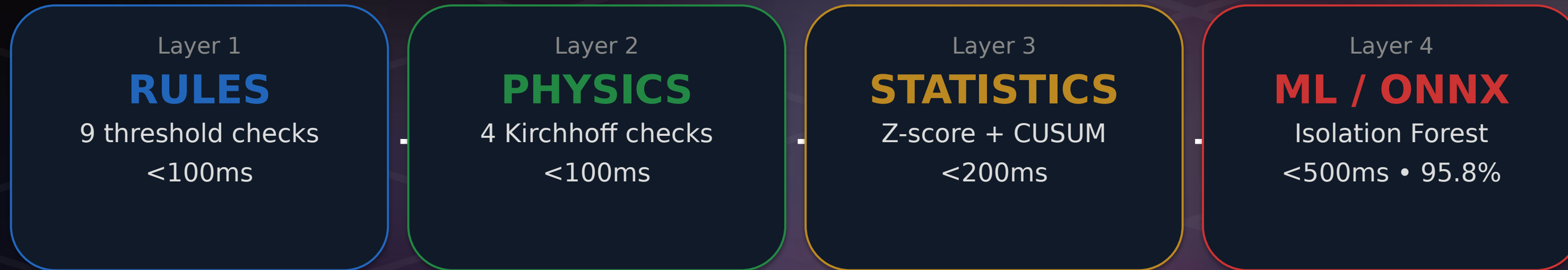
- Detection too slow — IT tools miss
physics-based FDI attacks
- Response is manual — 30-90 min to
isolate, every second = cascading risk
- No cyber-physical awareness — can't
tell attacks from legitimate faults
- AI underutilized — static rules easily
evaded by adversaries



SOLUTION



4-Layer AI Detection + Autonomous VajraShield™ FLISR Healing



ALERTCLASSIFIER FUSION → Confidence ≥ 0.90 → Triggers VajraShield™

VajraShield™ FLISR — Full Heal: 16 seconds (vs 30-90 min manual)

DETECT

0-3s

ISOLATE

3-5s

REROUTE

5-10s

MONITOR

10-13s

RESTORE

13-16s

DONE ✓

healed



SYSTEM ARCHITECTURE



PRESENTATION

Next.js 16 + React 19 | Tailwind v4 | Recharts + React Flow | Mobile SOC

API LAYER

REST: /api/simulation/{start,stop,attack} | SSE: /api/stream | Real-time push

SIMULATION

DataGenerator → 5-bus telemetry (V, f, P, Q, θ) | 5 Attack Injectors | Indian loads

DETECTION

Rules (9) → Physics (4) → Stats (Z+CUSUM) → ML/ONNX (Isolation Forest)

VAJRASHIELD™

FLISR: Locate → Isolate → Reroute → Restore | 16s full cycle

AMD HARDWARE

Edge: Ryzen AI NPU 50 TOPS + Vitis AI | Backend: EPYC AVX2+FMA3 | 15W

SCADA (1Hz) → API → DataGenerator → Detection → AlertClassifier (≥ 0.90) → VajraShield™



TECHNOLOGY USED



Frontend

Next.js 16 • React 19
TypeScript • Tailwind v4
Recharts • React Flow

AI / ML

ONNX Runtime
Isolation Forest
Vitis AI EP (edge)

Backend

Next.js API Routes
SSE streaming
Socket.IO real-time

Detection

Rule Engine (9 rules)
Physics (4 Kirchhoff)
Z-score + CUSUM

Simulation

5-bus grid model
5 attack injectors
Indian load curves

Self-Healing

VajraShield™ FLISR
AlertClassifier fusion
Confidence thresholds

Testing

Vitest: 37/37 pass
Playwright E2E
Testing Library

AMD Hardware

Ryzen AI NPU 50 TOPS
EPYC AVX2+FMA3
ROCm GPU training

Full working prototype • 37/37 tests passing • MIT Open Source • Not a concept



FEATURE / USP



⚡ **4-Layer Fusion**

Rules + Physics + Stats + ML
Each catches what others miss



Physics-Aware AI

Validates Kirchhoff's laws
Catches FDI invisible to IT



VajraShield™ FLISR

Autonomous healing in 16s
Zero human intervention



Edge AMD Ryzen AI

NPU: 50 TOPS, 15W, <5ms
Works on airgapped networks



Working Prototype

Live dashboard, 5 attacks
37/37 tests, real detection



Built for India

Indian load curves, 50Hz
Scales to 250K+ substations

Cyber-physical defense — NOT a generic cybersecurity tool



References / links



GitHub Repositories

Primary:

github.com/Imperialorg/vajragrid

Mirror:

github.com/Aerosane/vajragridr

- License: MIT Open Source
- Stack: Next.js 16 + ONNX
- Tests: 37/37 passing

Research References

- Dragos OT ICS IR Reports
- (2015-2024)
- IEEE 9-Bus Test System
- NERC CIP Standards
- ICS-CERT SCADA Advisories
- Insikt Group: RedEcho
- India Grid Targeting (2021)

AMD Documentation

- XDNA NPU Architecture
- Ryzen AI Developer Docs
- Vitis AI Execution Provider
- for ONNX Runtime
- EPYC AVX2/FMA3 Specs
- ROCm GPU Training Path

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THANK YOU

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github.com/Imperialorg/vajragrid